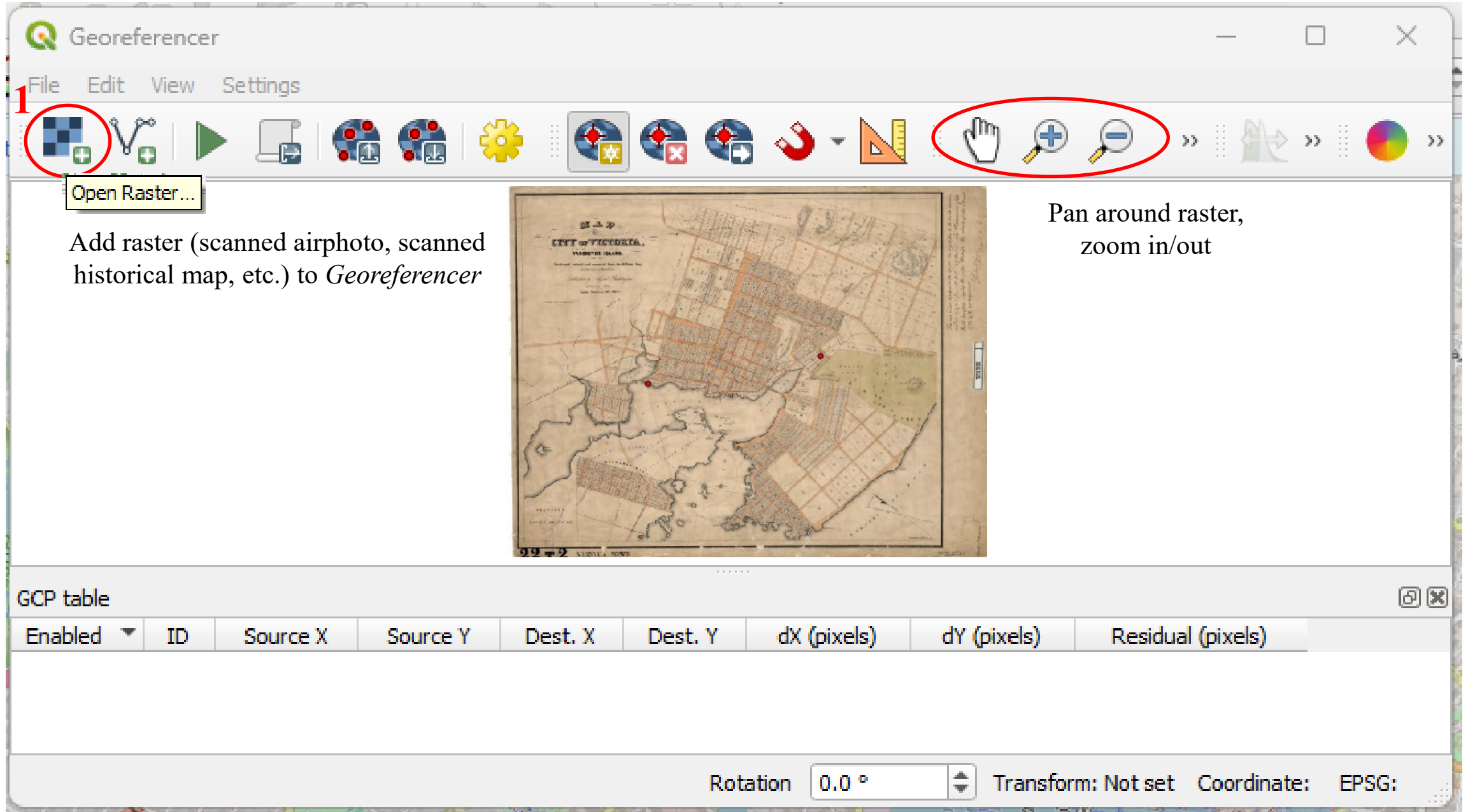


Georeferencing with *QGIS*

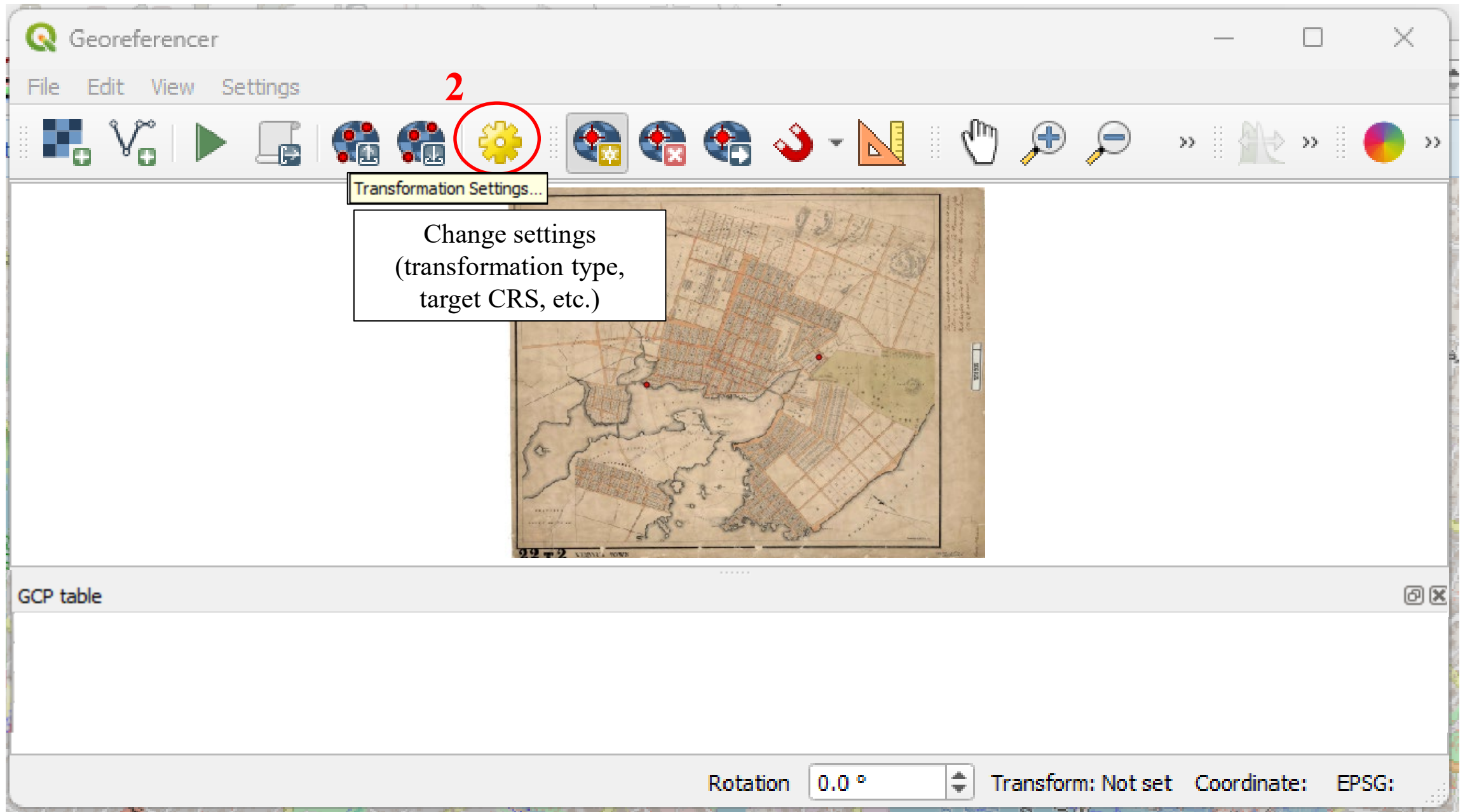


EXERCISE

Georeferencer interface



Georeferencer interface



The screenshot shows the Georeferencer application window. At the top, a text box says "Add ground control point to raster then basemap". Below it, the "Add GCP Point" button in the toolbar is circled in red, with a red "3" next to it. The main map area displays a historical map of Victoria, British Columbia, with two red dots indicating ground control points. A text box at the bottom left of the map area says "GCP points will appear here". At the bottom of the window, the "GCP table" is visible, containing two rows of data. The table has columns for Enabled, ID, Source X, Source Y, Dest. X, Dest. Y, dX (pixels), dY (pixels), and Residual (pixels).

Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
✓	0	1274.6662	-1822.9318	-13735388.97	6180446.31	0.000000	-0.000000	0.000000
✓	1	2873.5504	-1568.3323	-13732177.20	6177782.94	-0.000000	-0.000000	0.000000

Georeferencer interface

Once you have chosen GCPs and set transformation settings

File Edit View

Start Georeferencing

4



GCP table

Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
☒	0	1274.6662	-1822.9318	-13735388.97	6180446.31	0.000000	-0.000000	0.000000
☒	1	2873.5504	-1568.3323	-13732177.20	6177782.94	-0.000000	-0.000000	0.000000

Rotation 0.0 ° Transform: Not set Coordinate: EPSG:

Activity #1



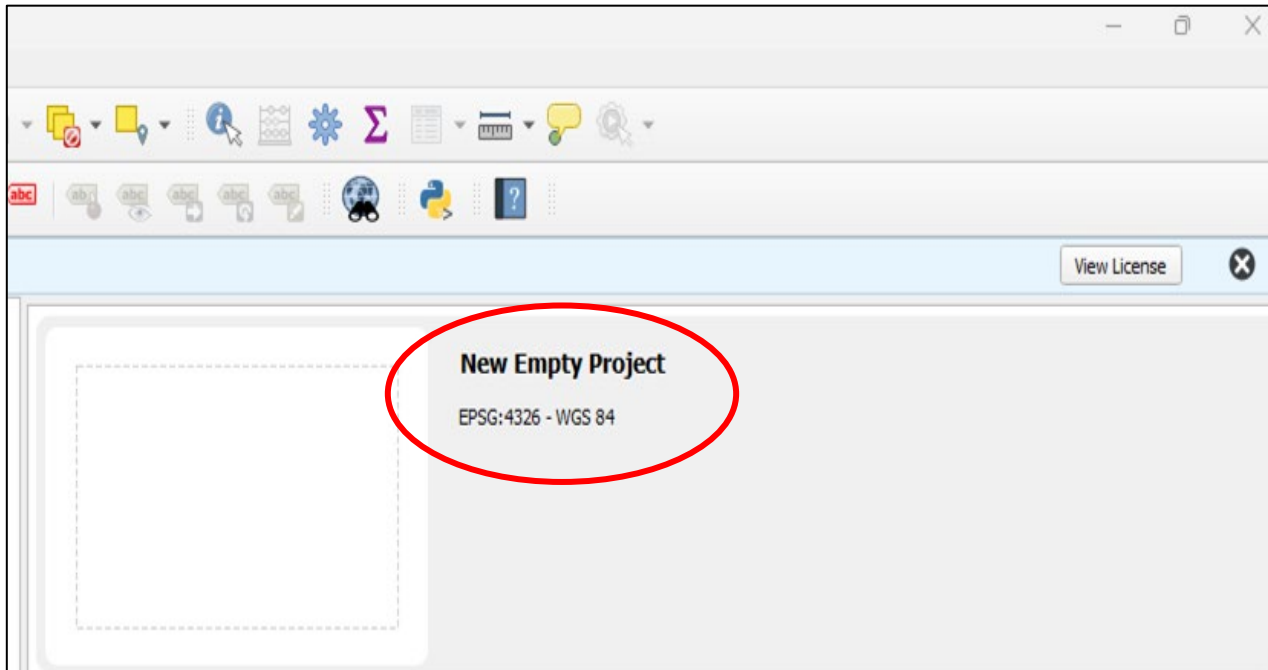
Start *QGIS*

Download workshop data

- Extract /unzip the .zip file
- Save it where you can find it...
- Can use either map provided, or your own scanned map as long as it is saved as a JPEG or TIFF file

Open *QGIS* (your version may be different)

- Double click on *New Empty Project*

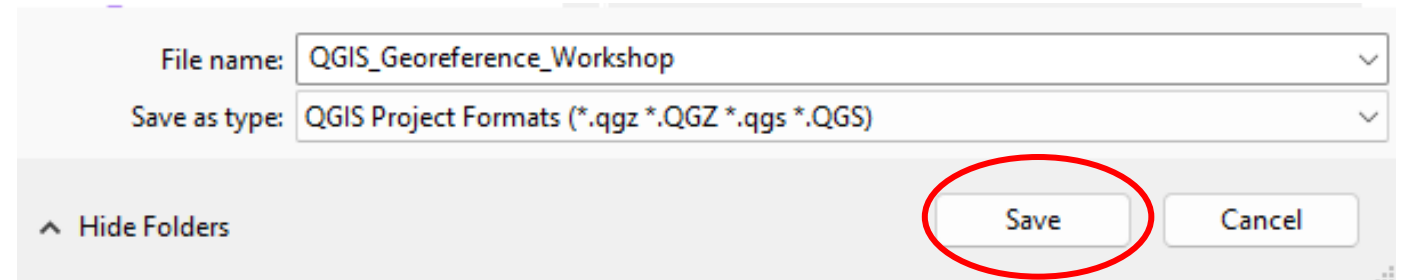
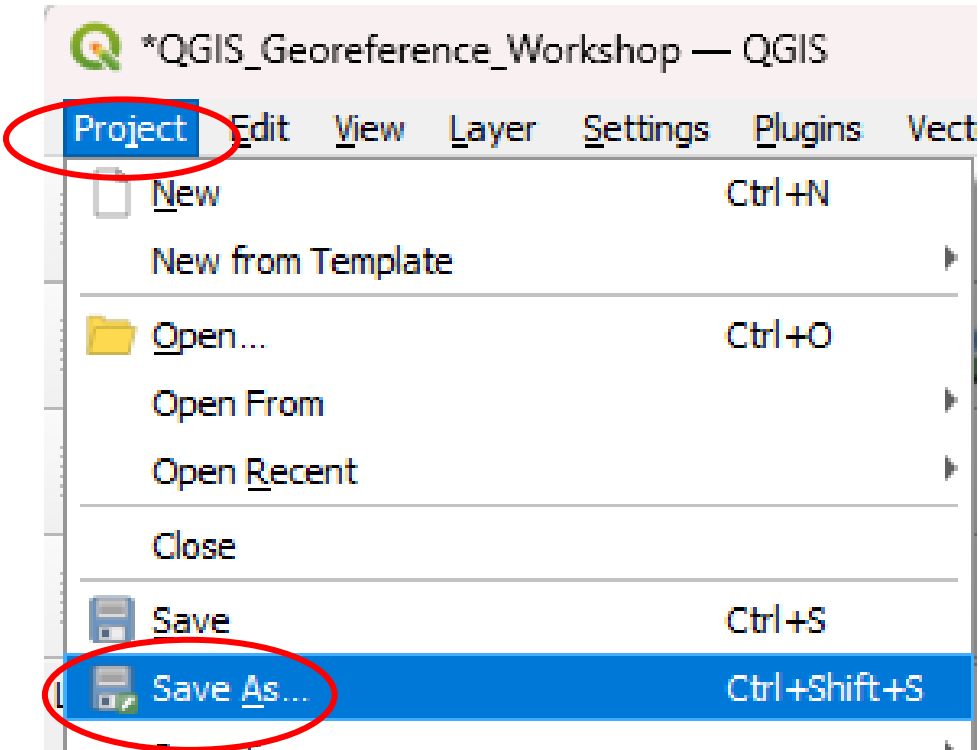


Note: New *QGIS* projects open with Geographic Coordinate System (GCS) **EPSG:4326**



Save new project

- In *QGIS* Menu Bar, select *Project* then *Save As*
- Name your project “**QGIS_Georeference_Workshop**”
- Save your project as **.qgz** to where you can find it



Note: .qgz is the project file format for *QGIS*

Add a Basemap

Add a **basemap** for location context

- In the 'Browser', expand *XYZ Tiles*
- Double-click *OpenStreetMap* to add to map (if a warning appears, press OK)
- The basemap will appear *Layers Panel* and *Map View*

The screenshot displays the QGIS interface with the following components:

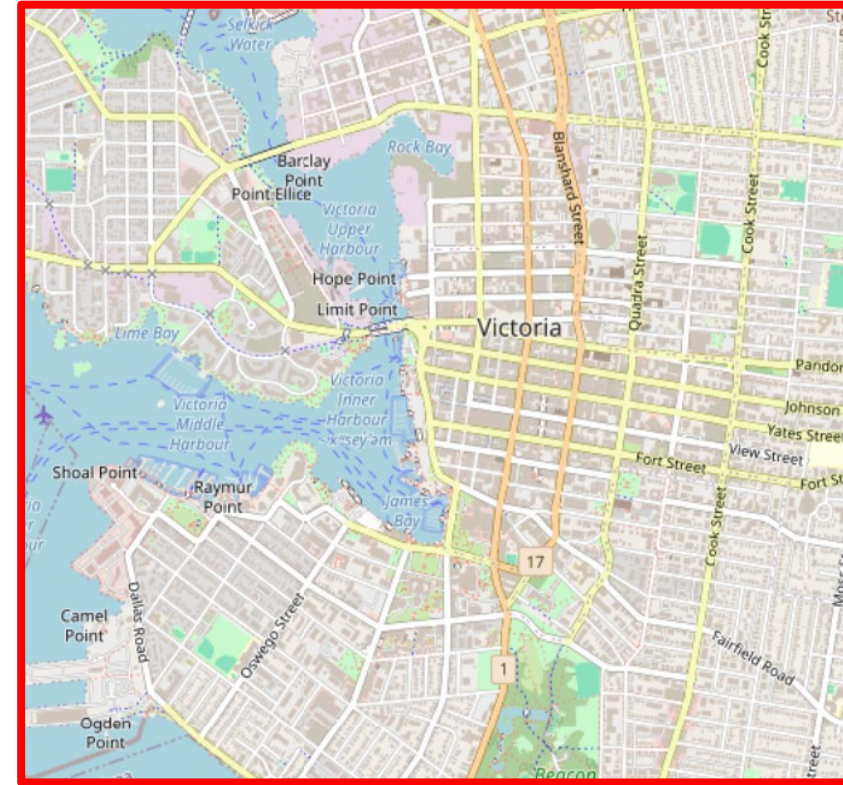
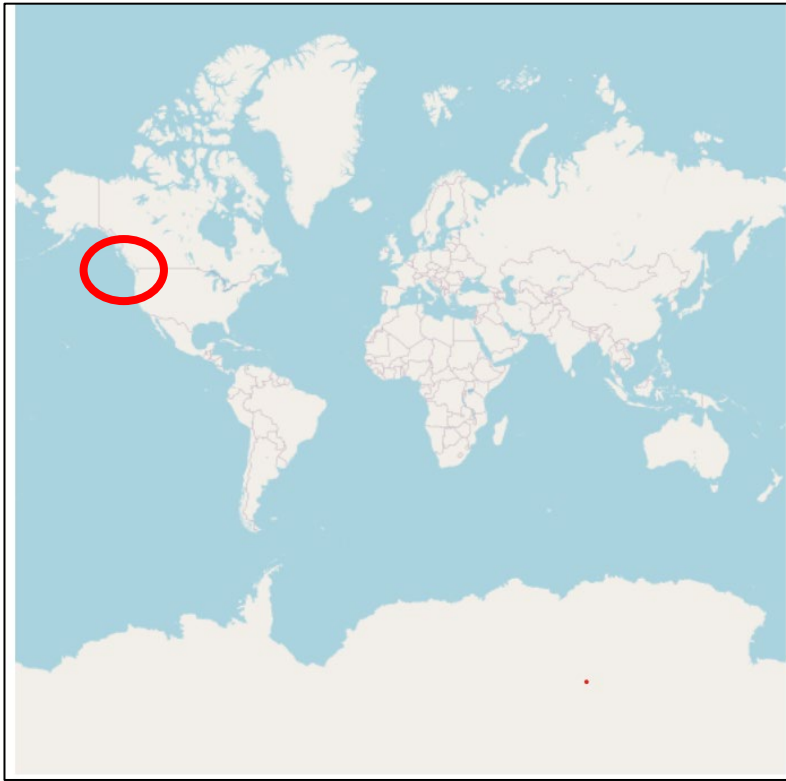
- Browser Panel (Left):** The 'XYZ Tiles' folder is expanded, and 'OpenStreetMap' is highlighted. Both are circled in red.
- Layers Panel (Top Left):** 'OpenStreetMap' is listed as the active layer.
- Map View (Center):** A world map showing the OpenStreetMap basemap.
- Project CRS Callout:** A text box states: 'Project CRS automatically changed to CRS of **OpenStreetMap** (EPSG: 3857)'.
- CRS Callout:** A red-bordered box shows the globe icon and 'EPSG:3857'.
- Status Bar (Bottom):** The coordinate system is listed as 'EPSG:3857'.

Zoom to location

- Using scroll wheel or Zoom tool

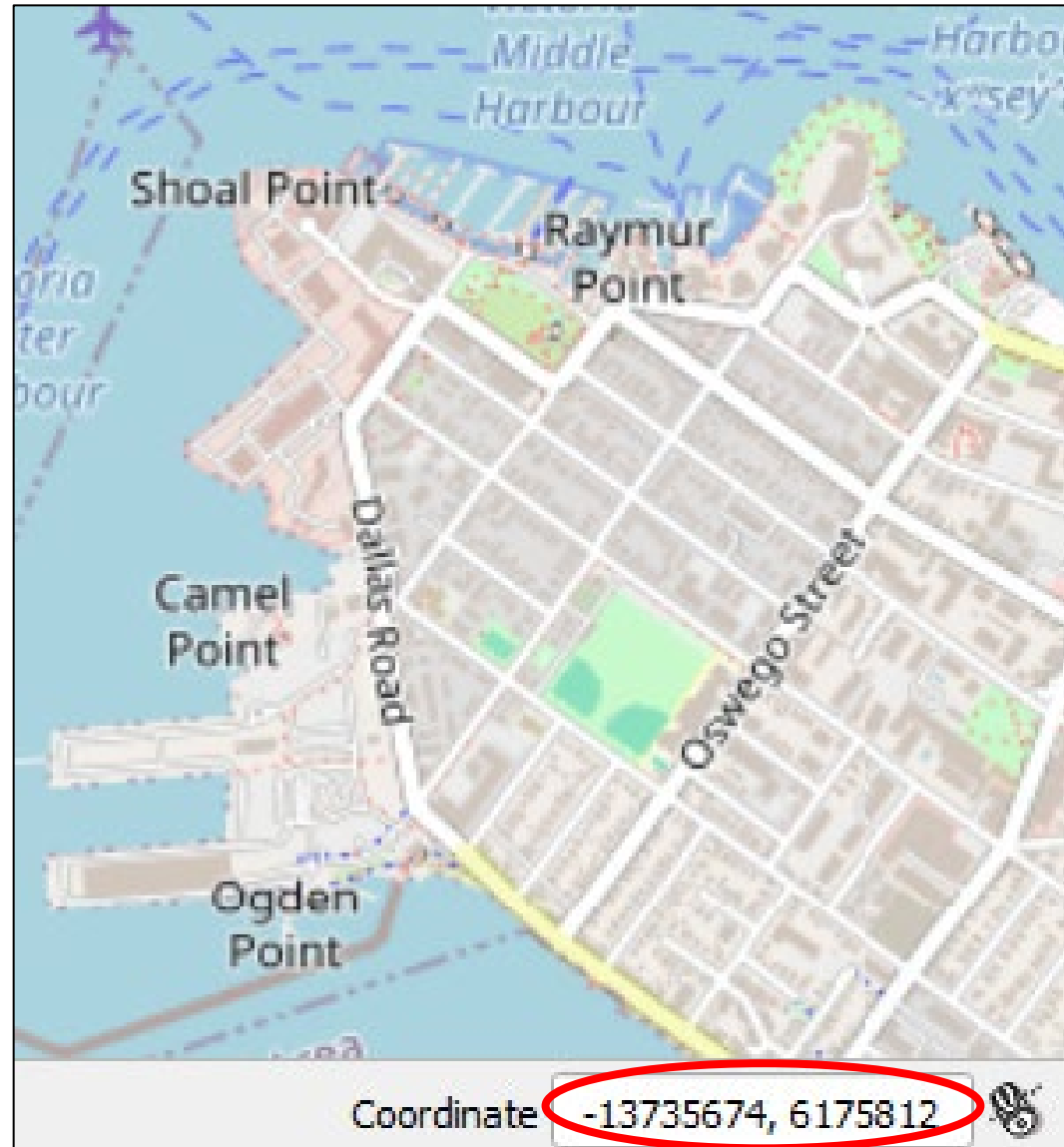


to zoom into study area (Victoria, BC)



Change coordinate settings

QGIS default setting for *Coordinate* is in metres



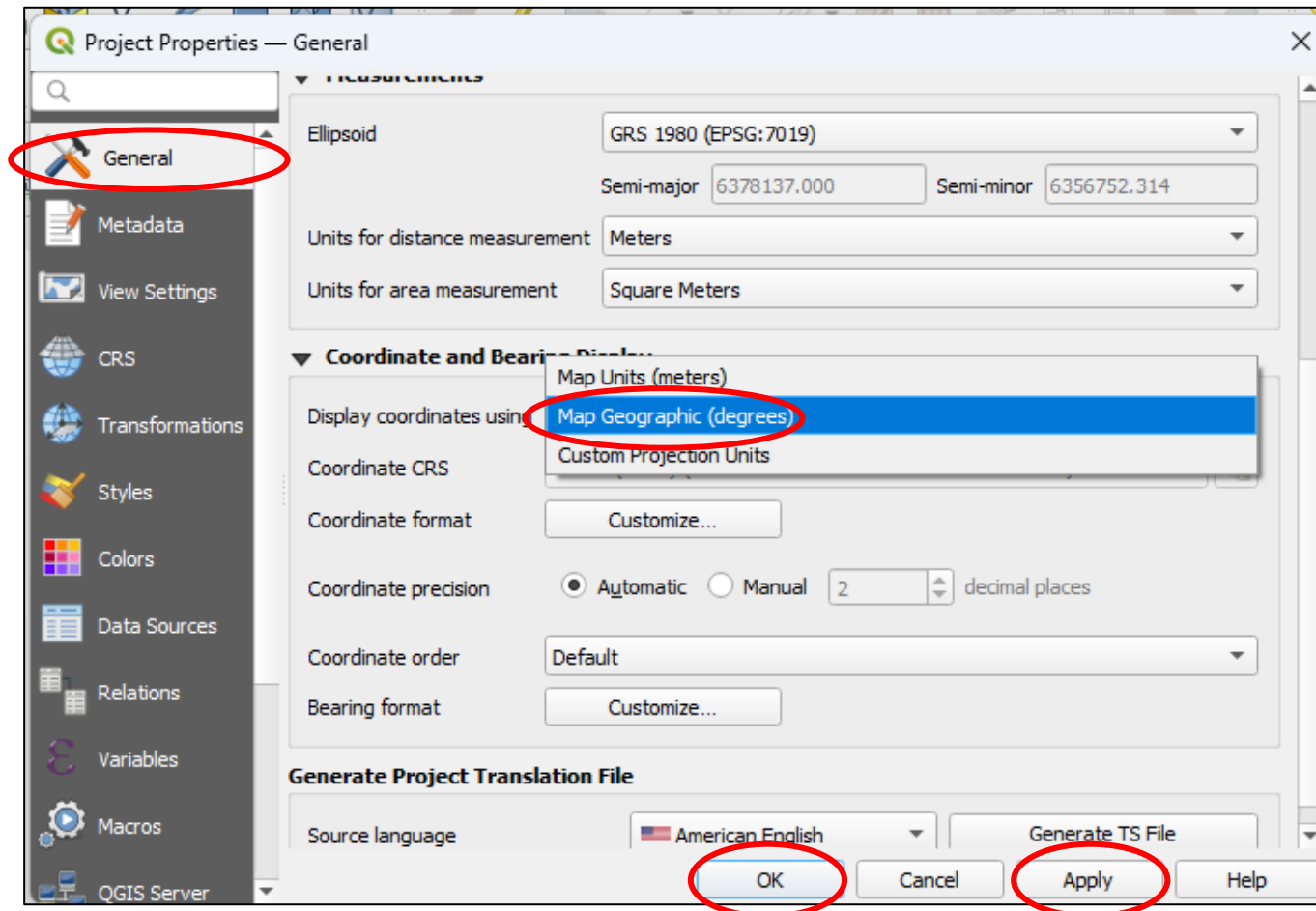
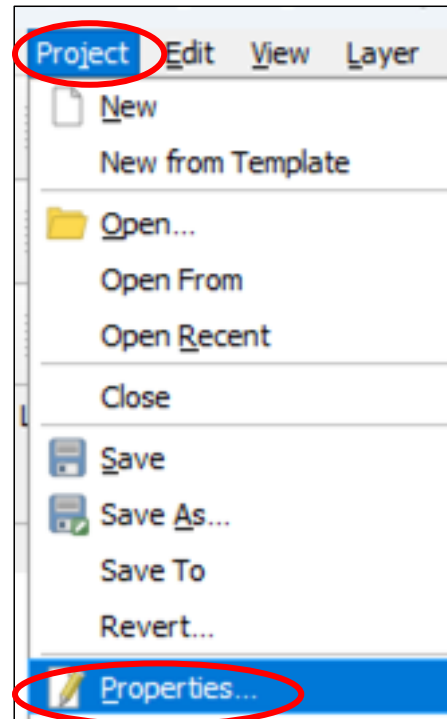
Change coordinate settings

Change coordinate settings from metres
to decimal degrees

Coordinate 451519, 5382443

Coordinate -123.6154°, 48.5755°

- In the *Menu Bar*, select *Project* then *Properties*
- In the *General* tab, scroll down to *Coordinate and Bearing Display* and use the drop-down to change *Display coordinates using* to **Map Geographic (degrees)**
- **Apply** and **OK**



Coordinate -123.6154°, 48.5755°

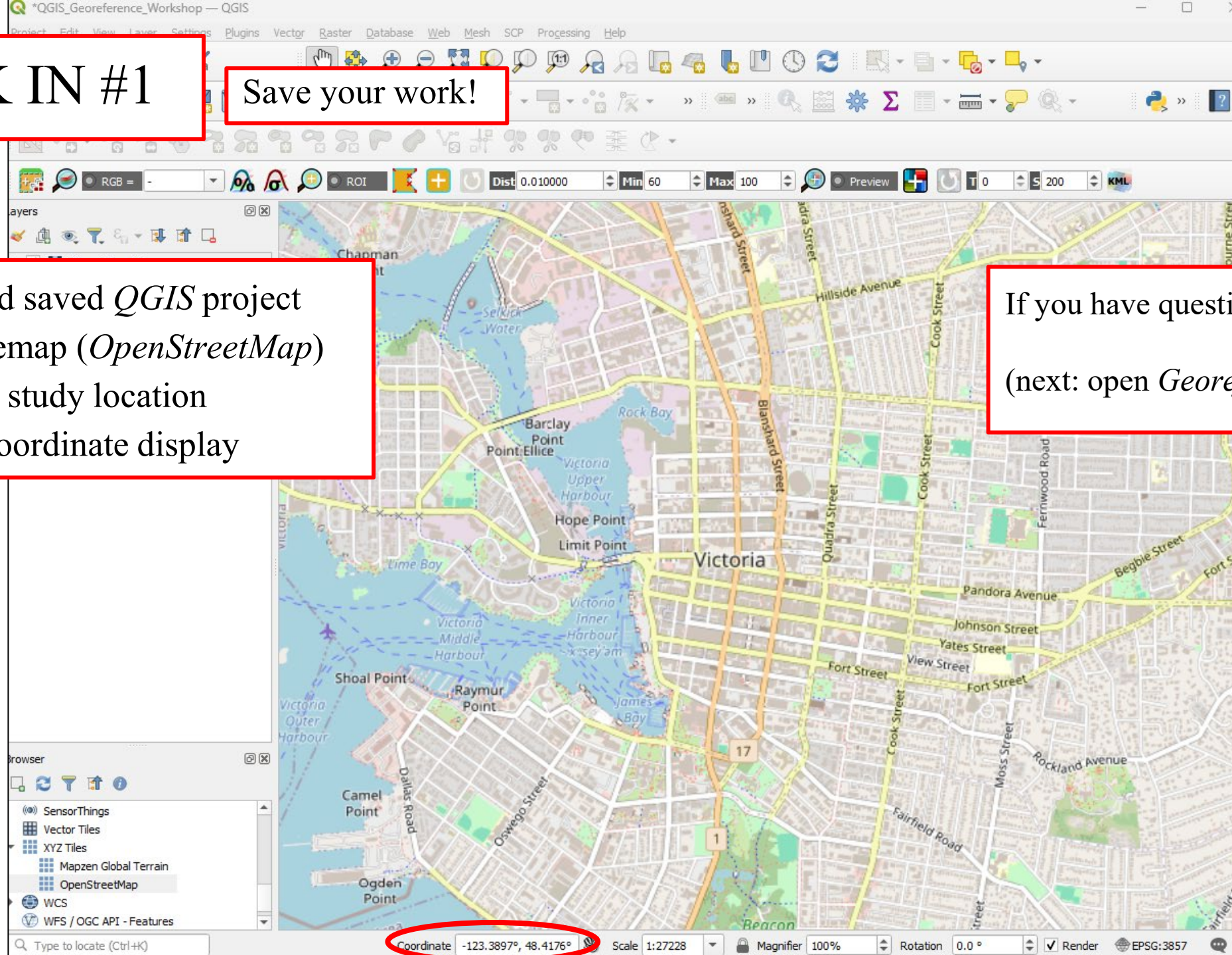


CHECK IN #1

Save your work!

- Opened and saved *QGIS* project
- added Basemap (*OpenStreetMap*)
- Zoomed to study location
- Changed coordinate display

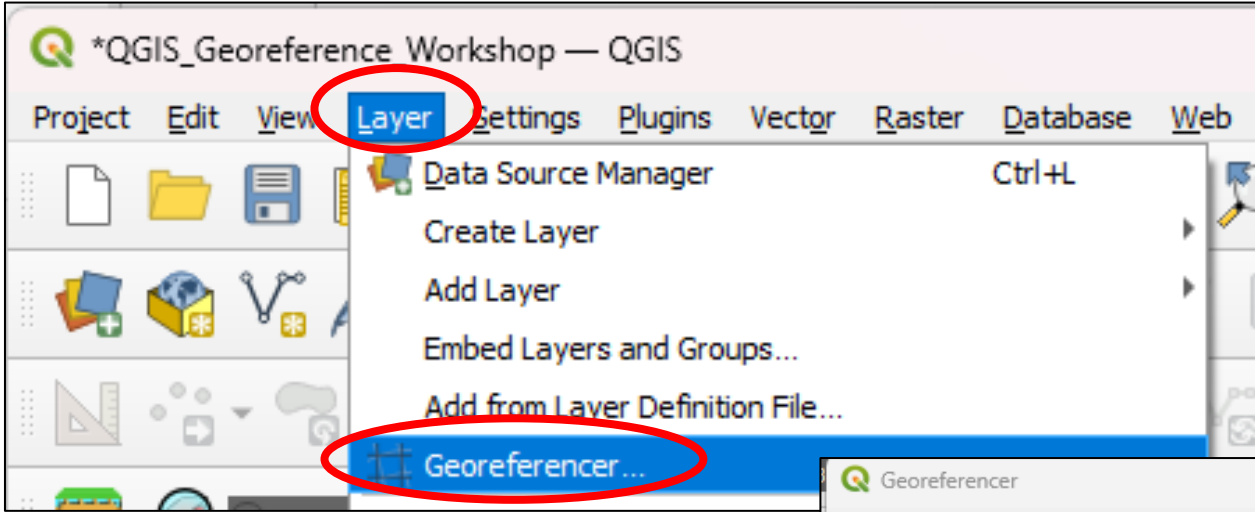
If you have questions, **ask!**
(next: open *Georeferencer* tool)



Activity #2

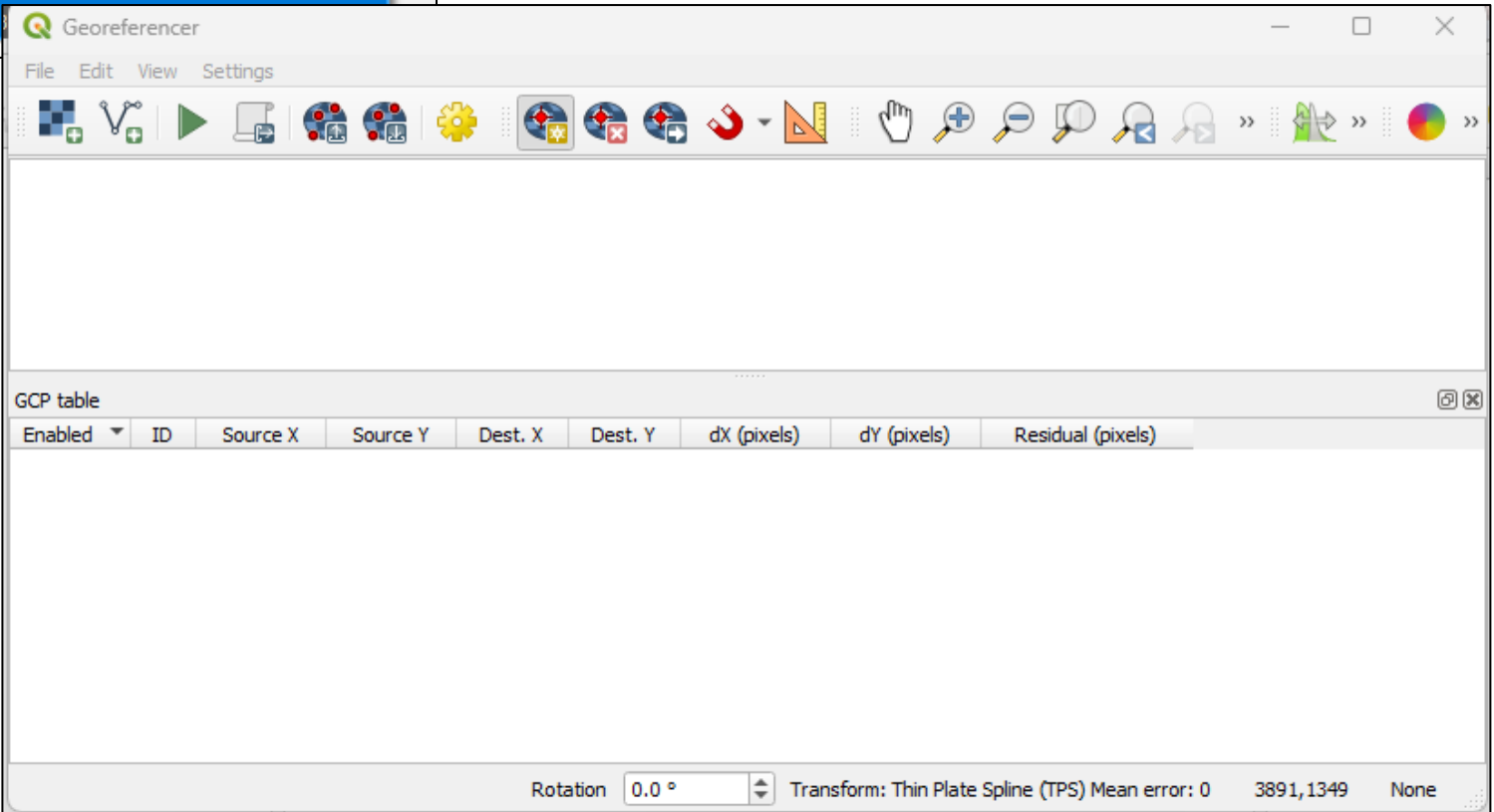


Open Georeferencer Tool



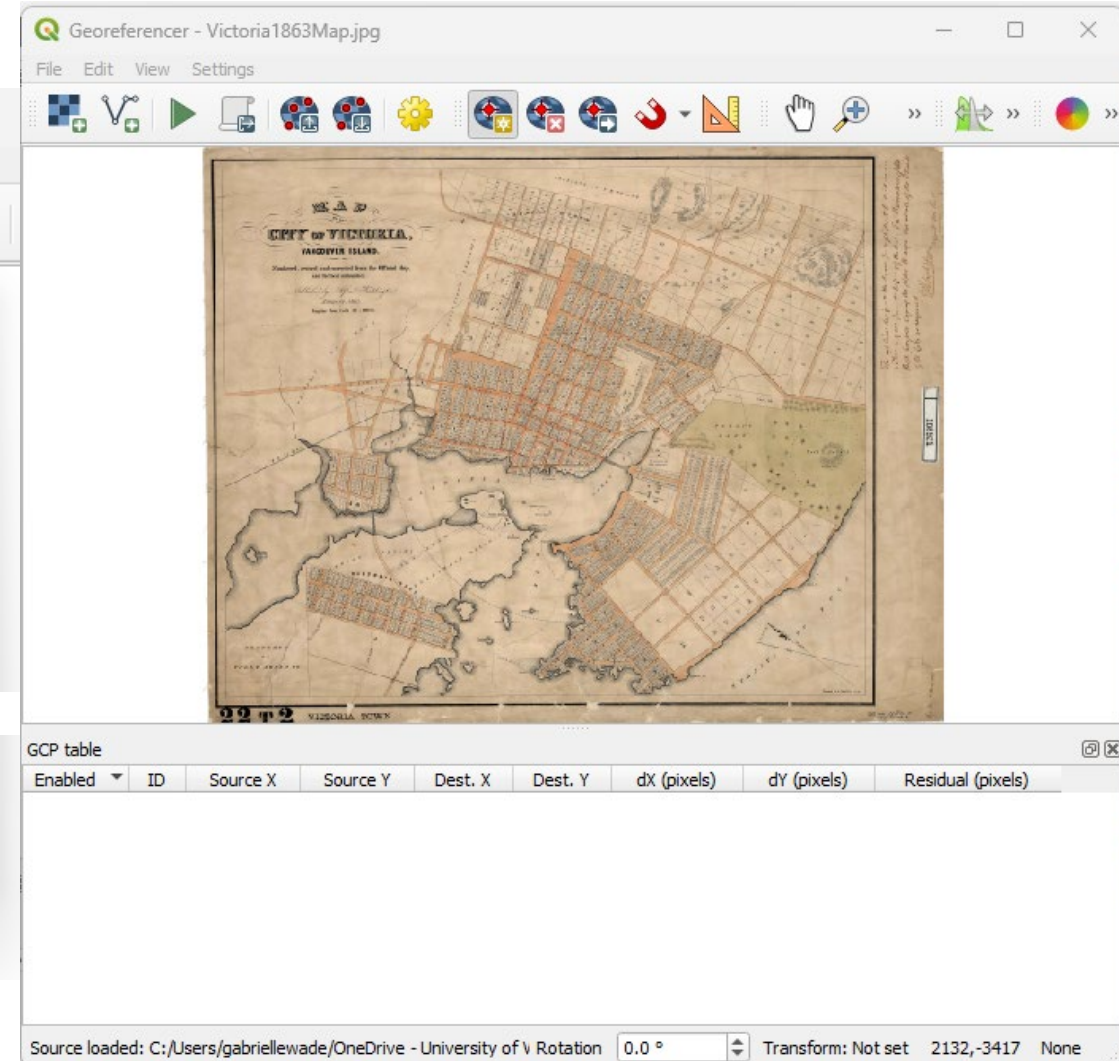
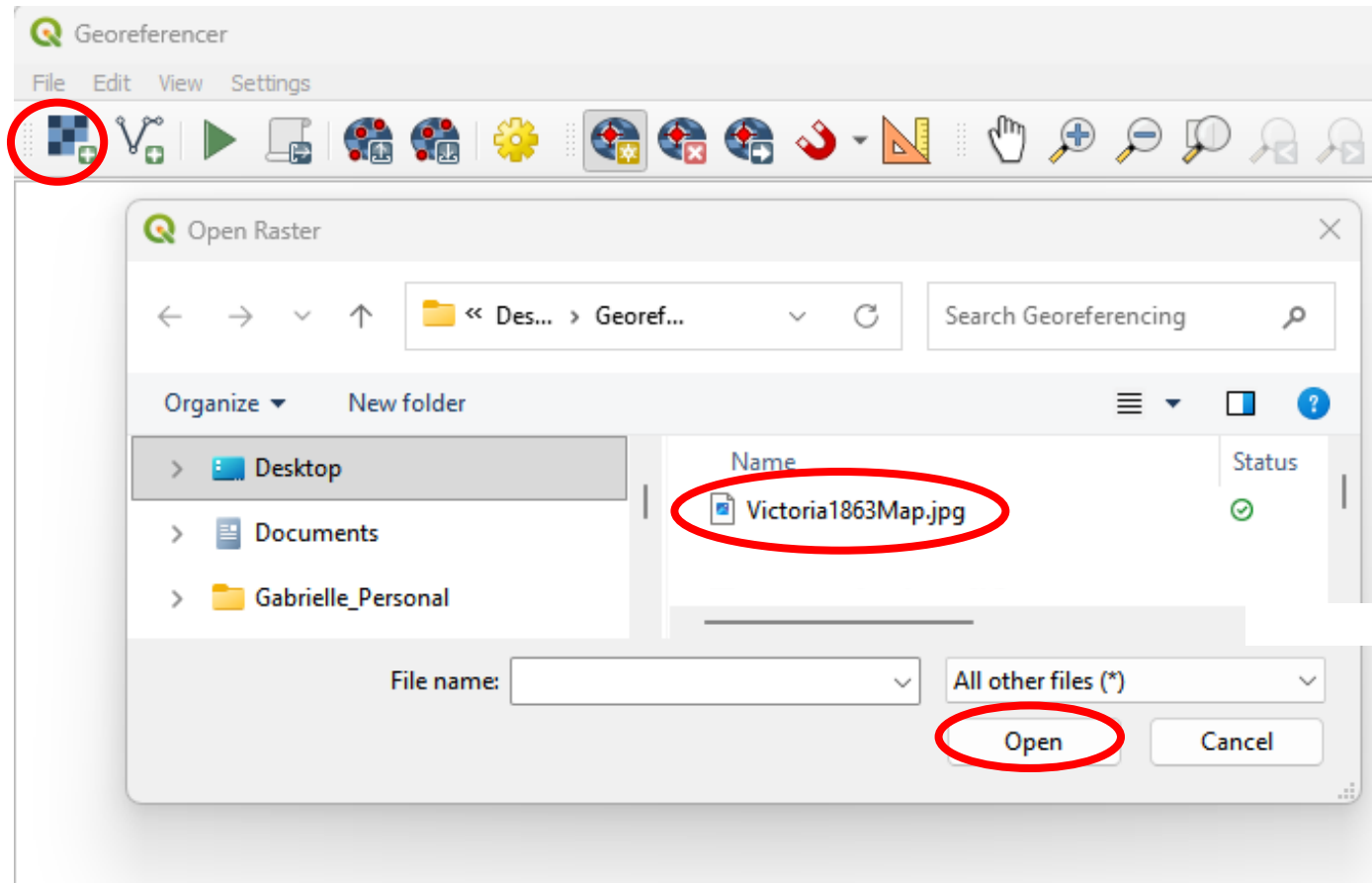
- In Menu Bar, select *Layer* and then *Georeferencer*
- Georeferencer window will pop up

Georeferencer window →



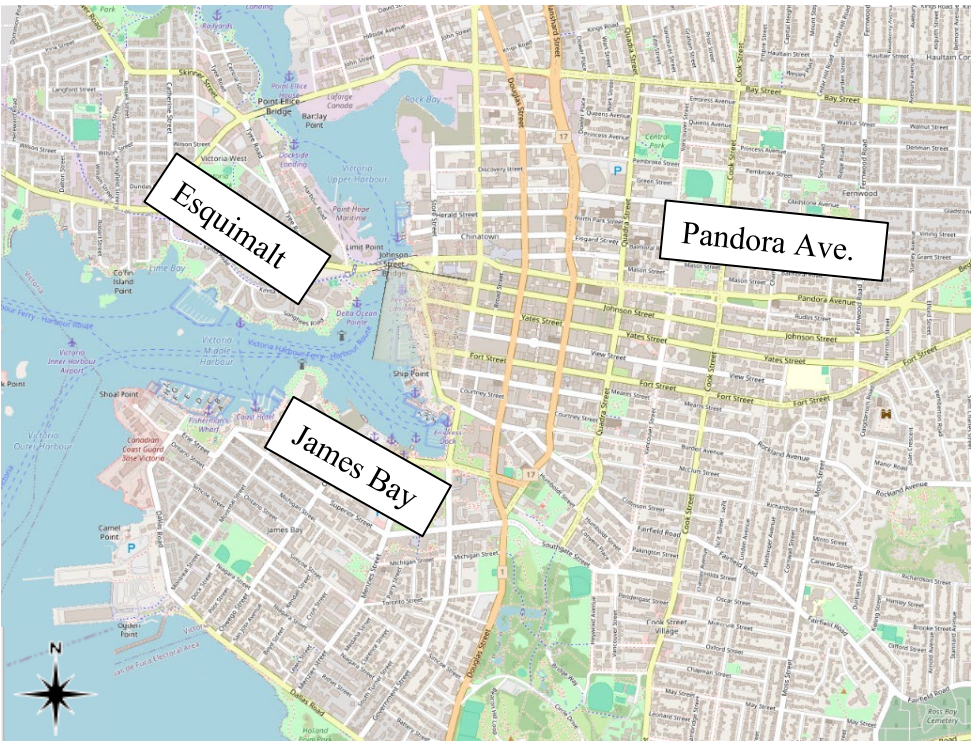
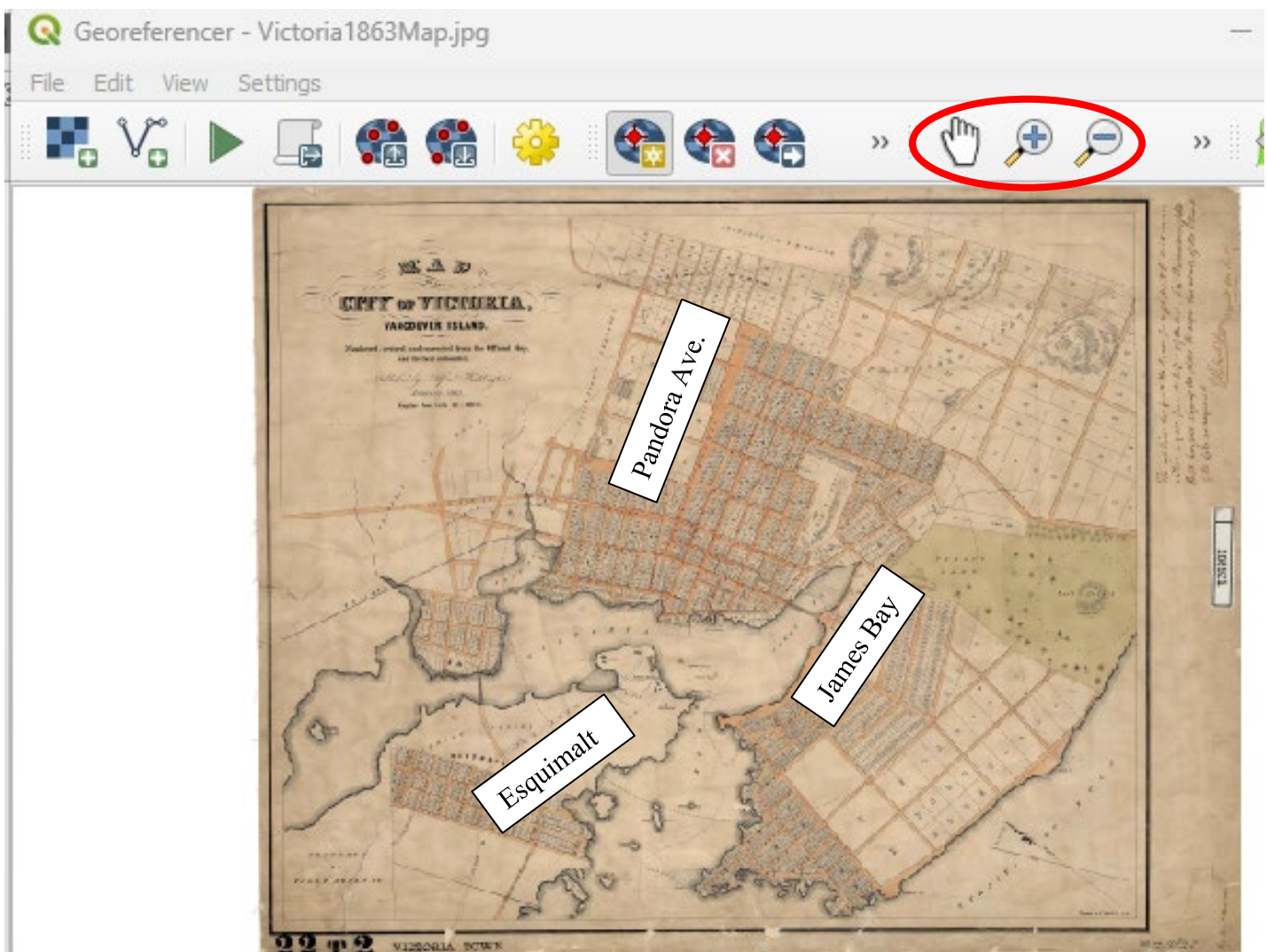
Add map

- In Georeferencer, click *Open Raster* 
- Navigate to workshop folder, select **Victoria1863Map.jpg** and then *Open*
- Map will now be in top section of Georeferencer pane



Explore historical map

- Using the Pan and Zoom    tools, explore the historical map
- Try to determine familiar locations and features to match to basemap



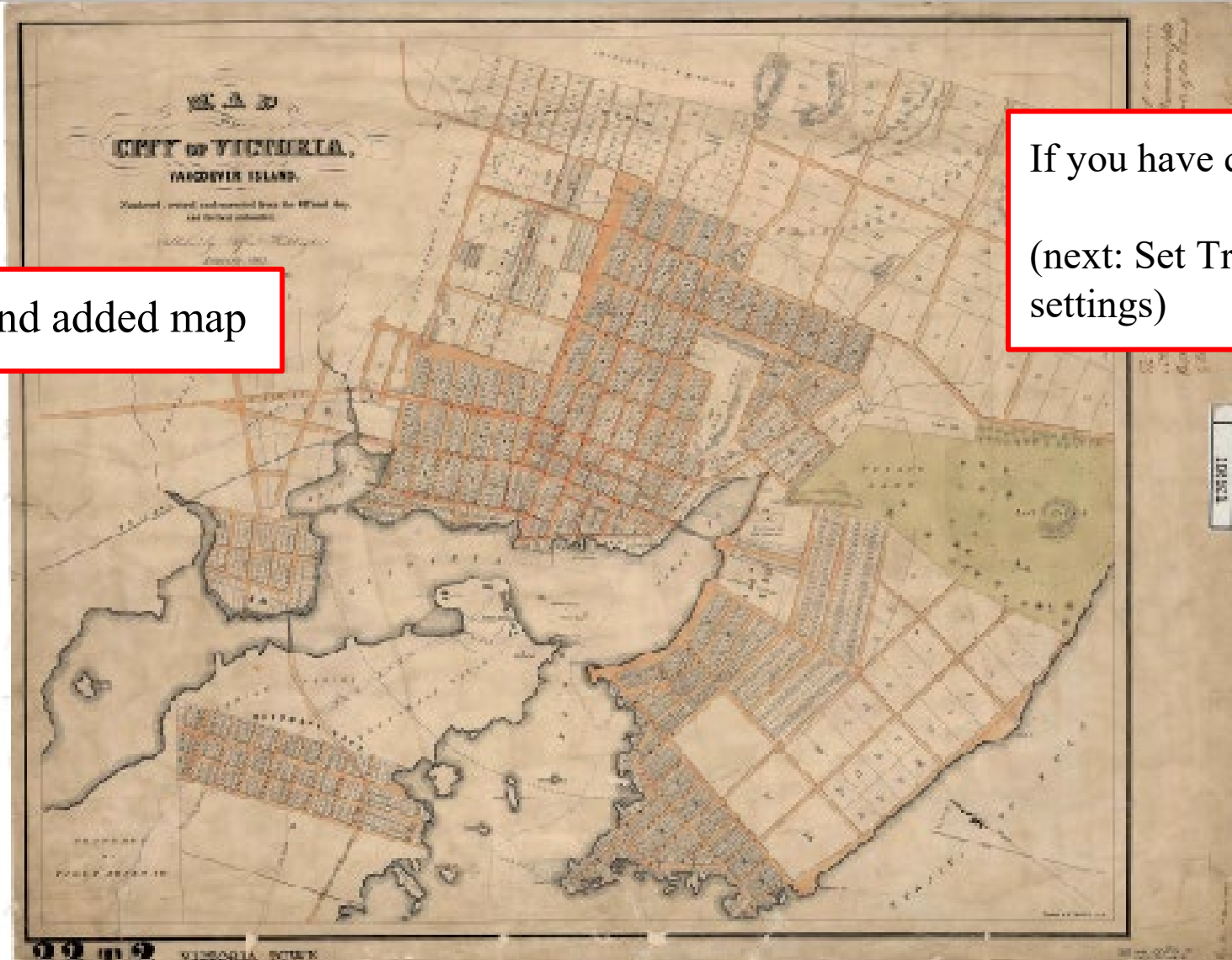
CHECK IN #2

Save your work!

- Opened Georeferencer and added map

If you have questions, **ask!**

(next: Set Transformation settings)

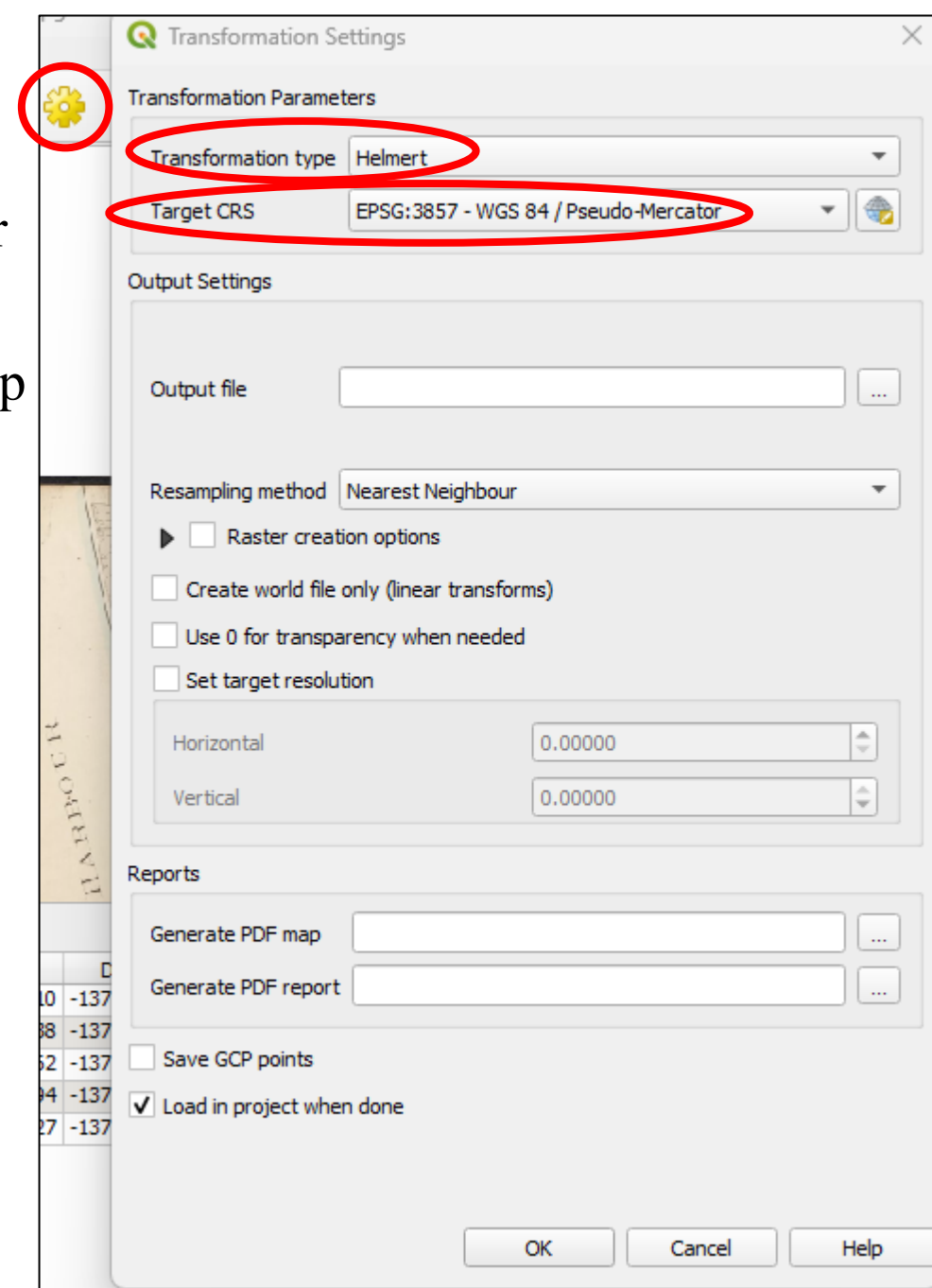


Activity #3



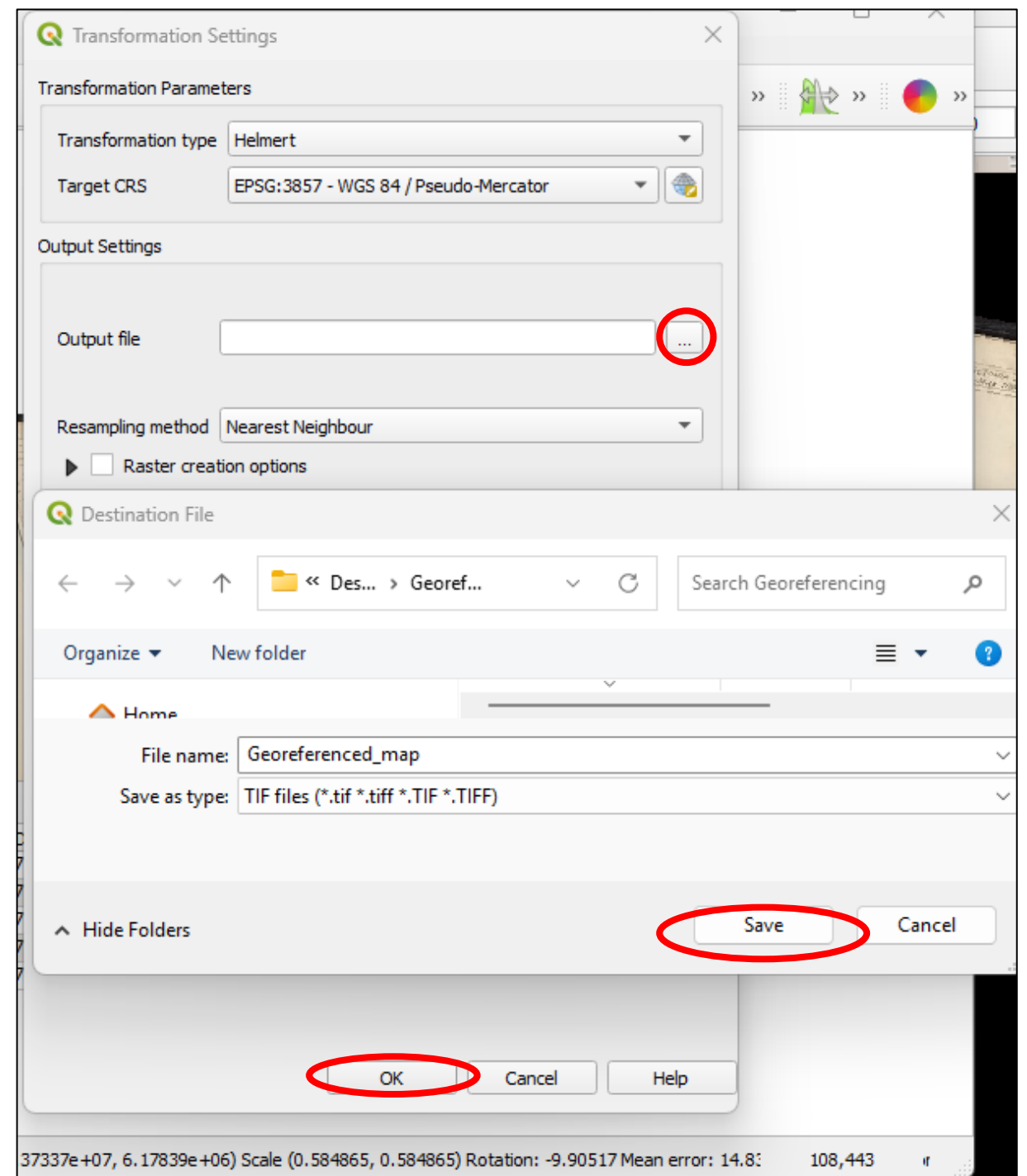
Transformation settings

- Select *Transformation Settings*  in the *Georeferencer* toolbar
- Select “Transformation type” as *Helmert*
- Ensure “Target CRS” is *EPSG:3857* to match CRS of basemap



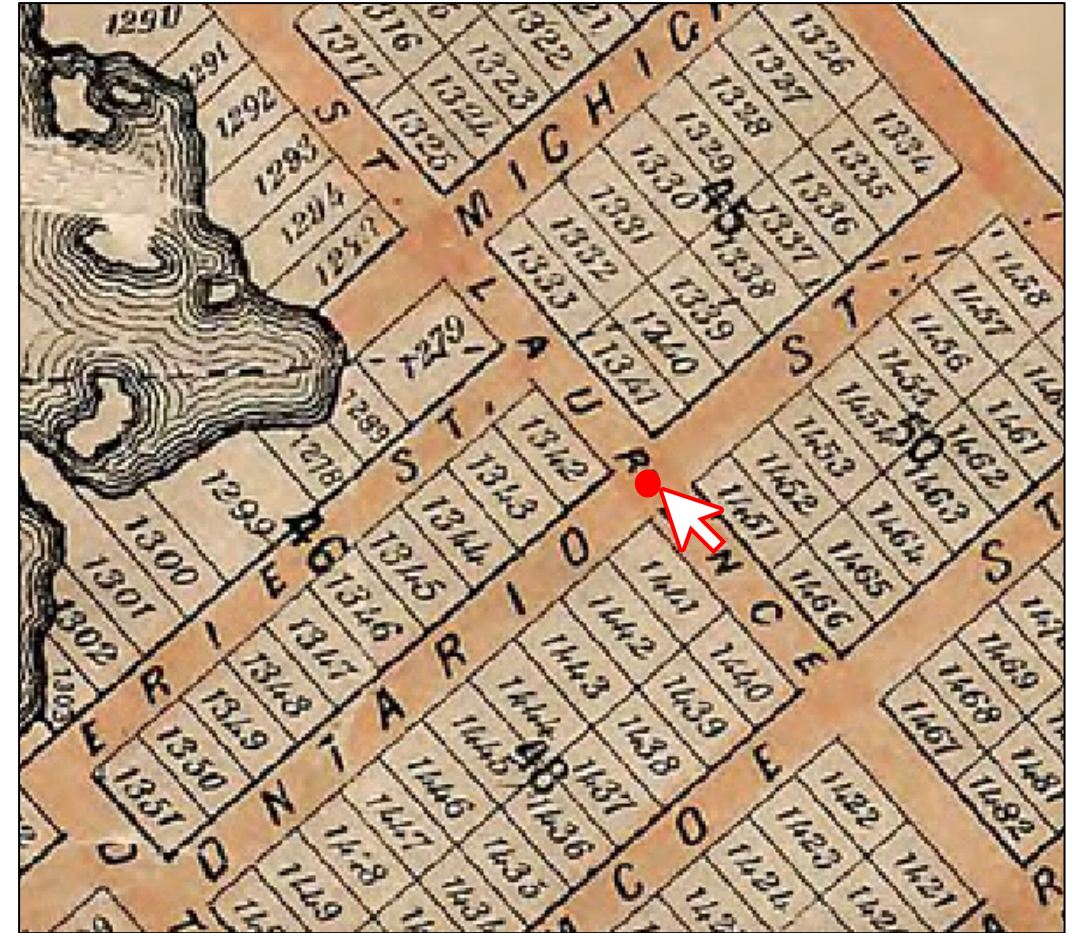
Transformation settings

- Select 3 dots beside *Output file*, navigate to workshop folder, and save file as “Georeferenced_map”
- Click *OK*



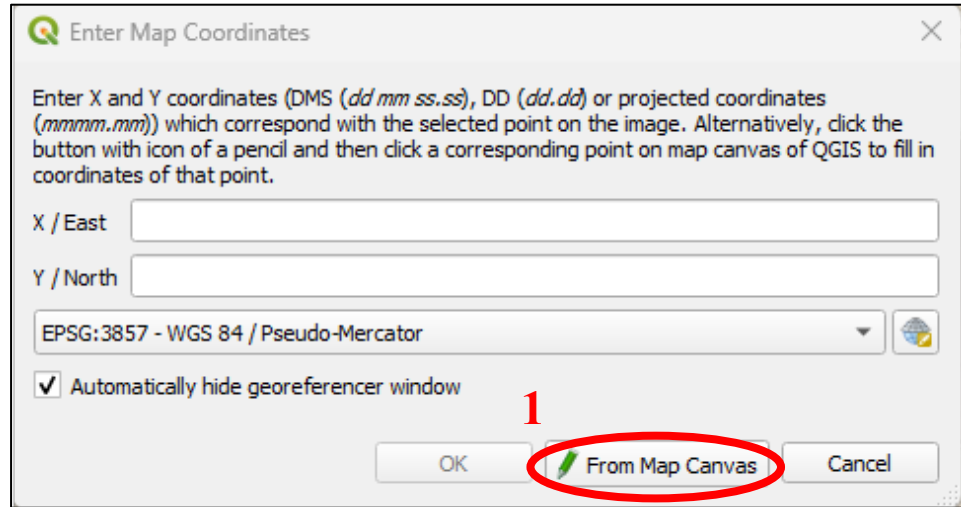
Add Ground Control Points

- Will use GCPs to link locations on map to spatially referenced target data (**OpenStreetMaps**)
- Select *Add GCP Point* tool 
- Select GCP: example given below (intersection of Ontario St. and St. Laurence/Lawrence St.)



Add Ground Control Points

- *Enter Map Coordinates* pop-up window will appear, select *From Map Canvas*
- Select same location on **OpenStreetMaps**
- Click *OK* in *Enter Map Coordinates* window



Enter Map Coordinates

Enter X and Y coordinates (DMS (*dd mm ss.ss*), DD (*dd.dd*) or projected coordinates (*mmmm.mm*)) which correspond with the selected point on the image. Alternatively, click the button with icon of a pencil and then click a corresponding point on map canvas of QGIS to fill in coordinates of that point.

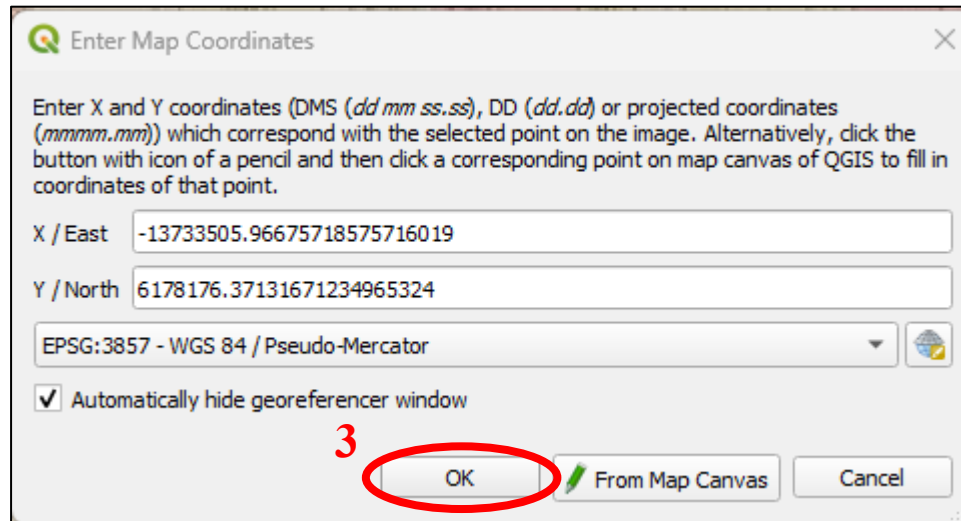
X / East

Y / North

EPSG:3857 - WGS 84 / Pseudo-Mercator

☒ Automatically hide georeferencer window

OK **From Map Canvas** Cancel



Enter Map Coordinates

Enter X and Y coordinates (DMS (*dd mm ss.ss*), DD (*dd.dd*) or projected coordinates (*mmmm.mm*)) which correspond with the selected point on the image. Alternatively, click the button with icon of a pencil and then click a corresponding point on map canvas of QGIS to fill in coordinates of that point.

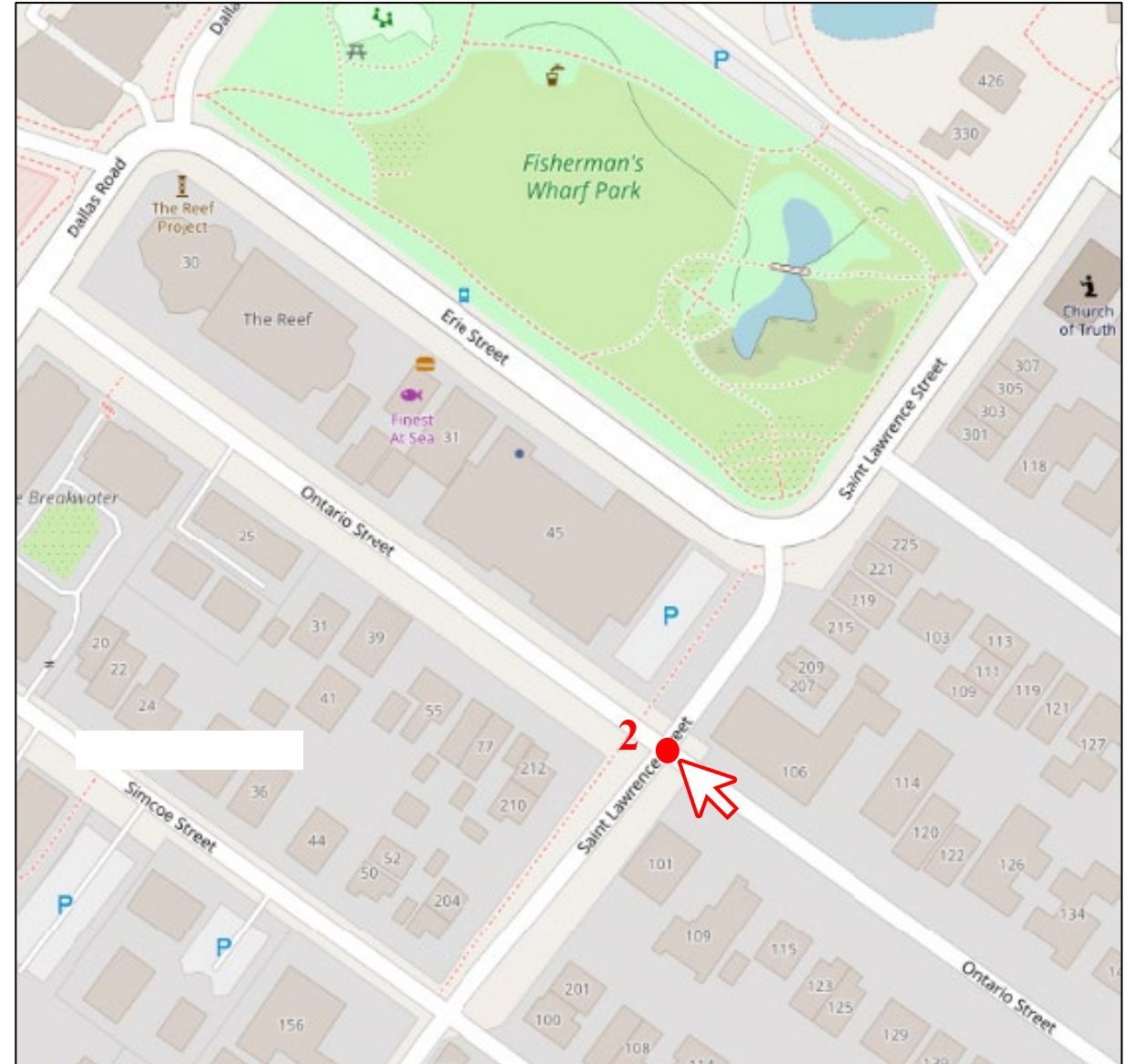
X / East -13733505.96675718575716019

Y / North 6178176.37131671234965324

EPSG:3857 - WGS 84 / Pseudo-Mercator

☒ Automatically hide georeferencer window

OK From Map Canvas Cancel



Choose Ground Control Points

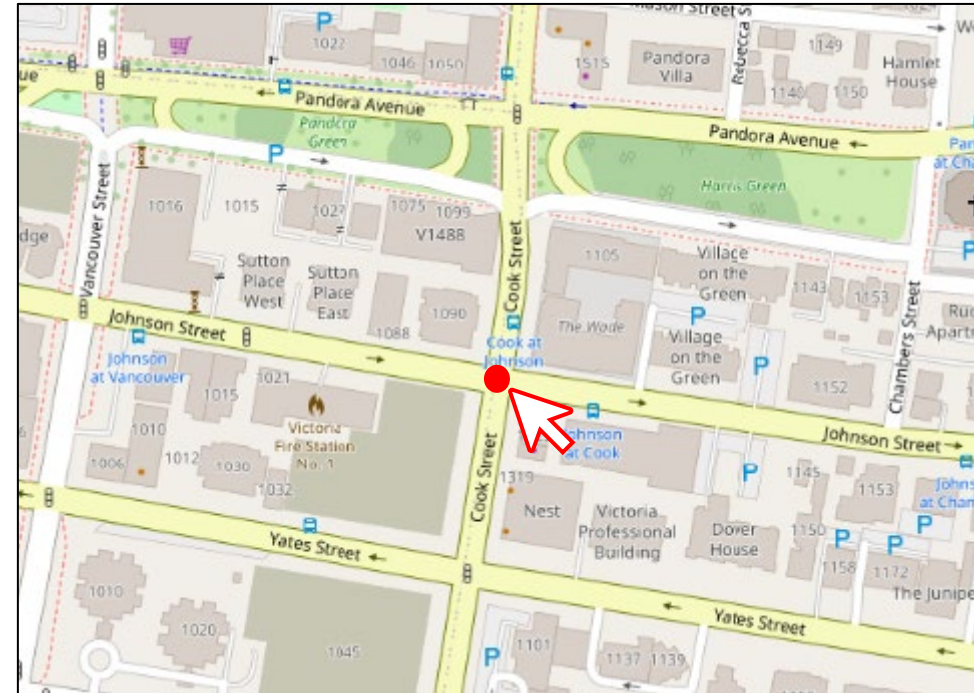
- New GCP will show up in *GCP table* in Georeferencer window



GCP table								
Enabled ▾	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
☒	0	432.874679	-300.201610	-13733505.97	6178176.37	n/a	n/a	n/a

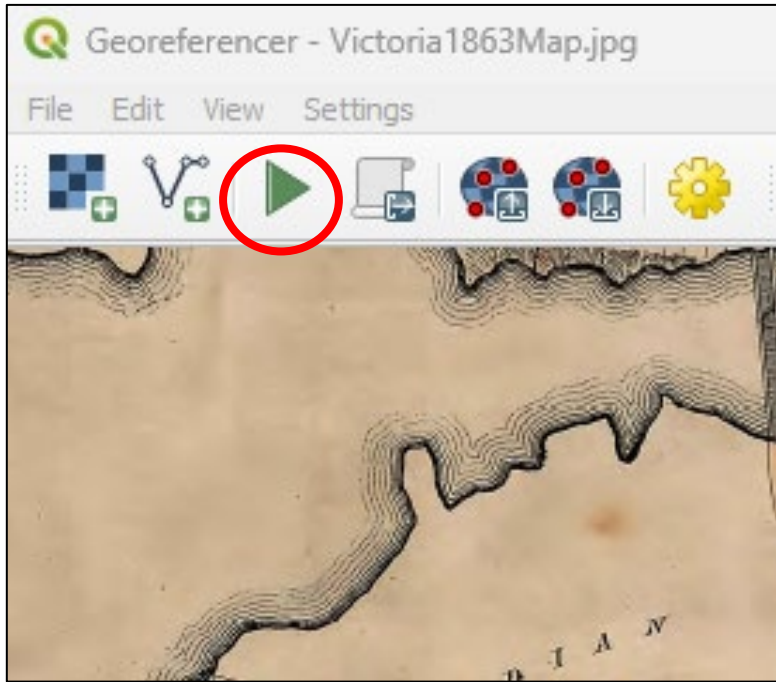
- Select **at least 1 more** GCP, try to be as accurate as possible and spread the points throughout the map
 - Remember: *Helmert* transformation requires at least **2** GCPs
 - Example of second point on next page (intersection of Johnson and Cook)

Choose Ground Control Points



Once at least 2 GCPs chosen, can Start Georeferencing

Start georeferencing



- In the Georeferencer window, click the play button to *Start georeferencing*
- Georeferenced map will appear in the *Map View* and *Layers*
- Minimize Georeferencer window



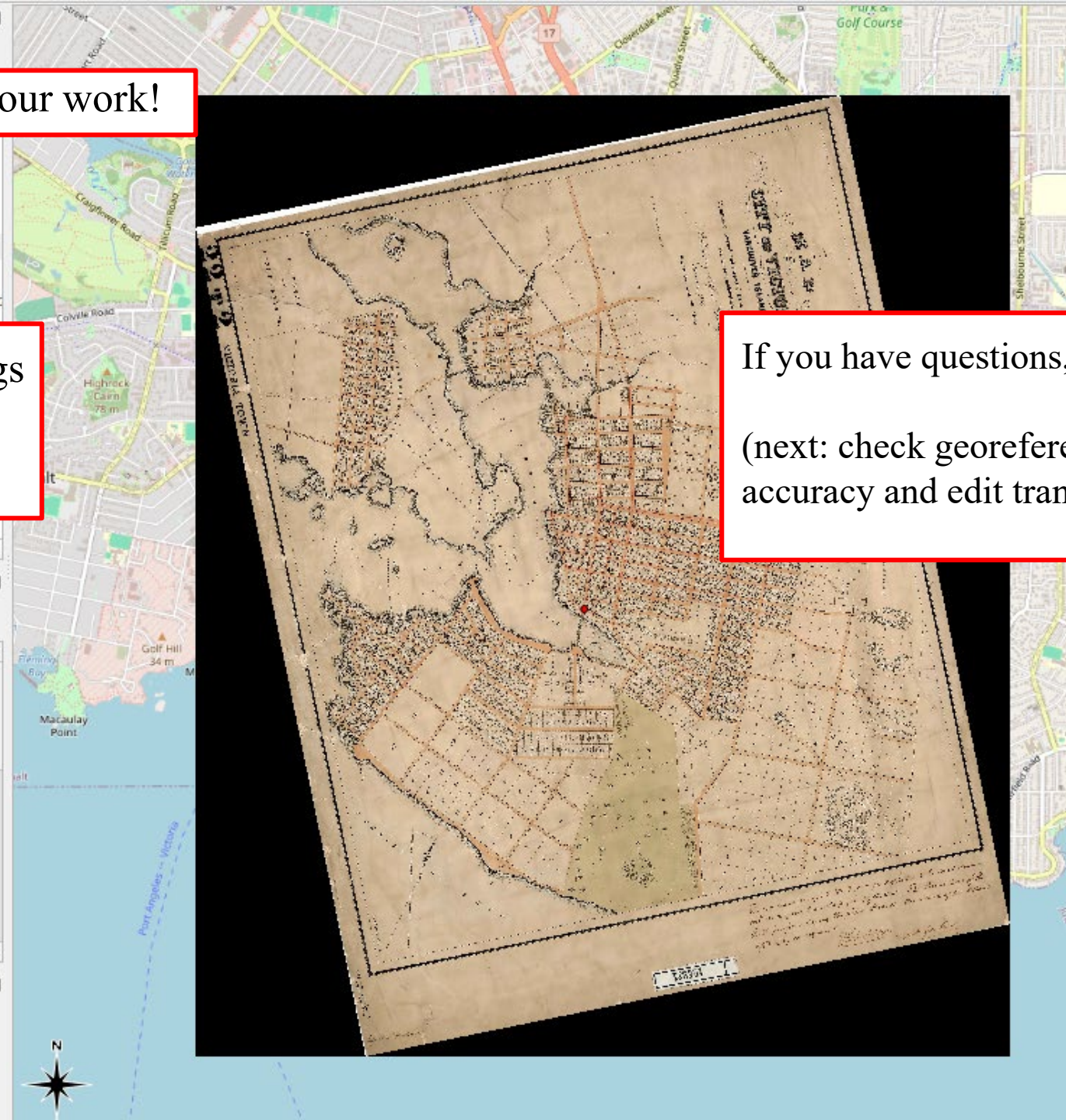
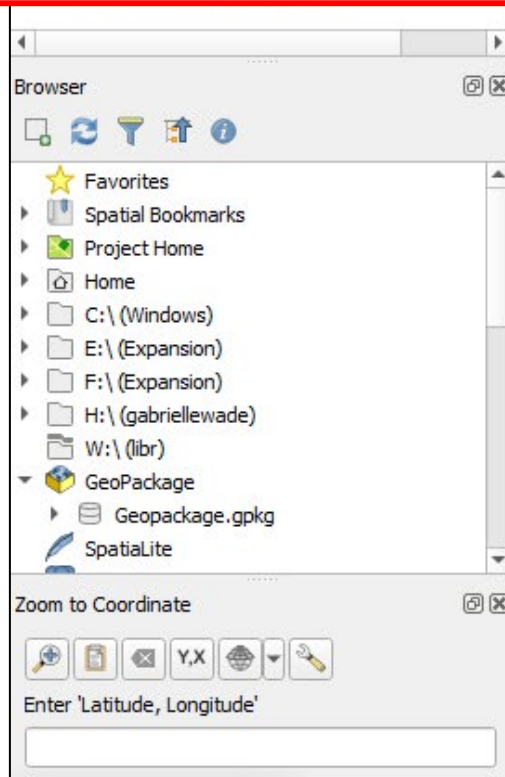
CHECK IN #3

Save your work!

- Customized Transformation settings
- Chose Ground Control Points
- Georeferenced map

If you have questions, **ask!**

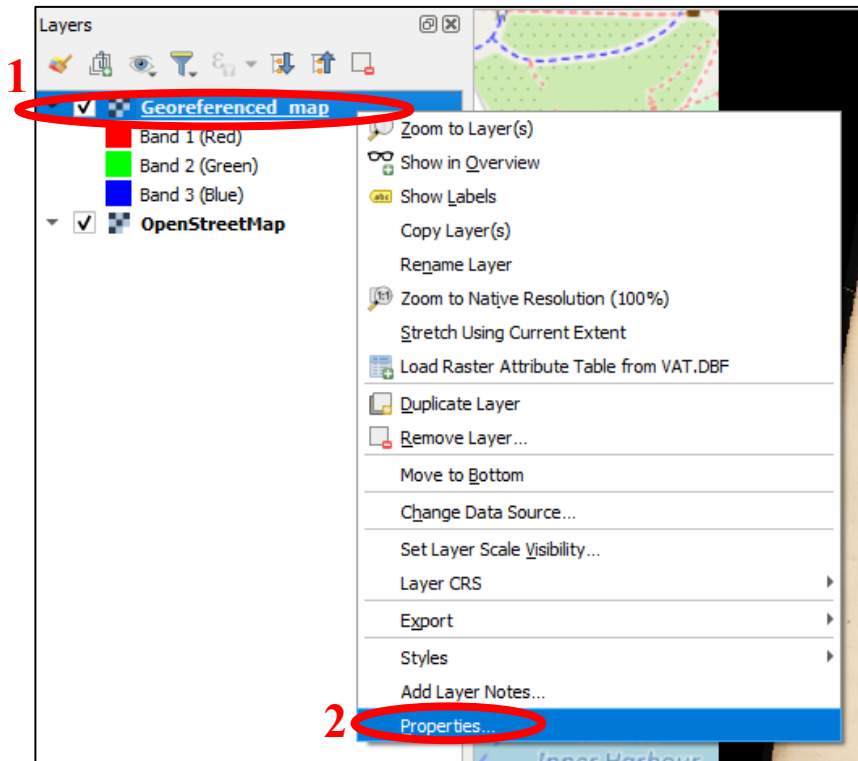
(next: check georeferencing accuracy and edit transparency)



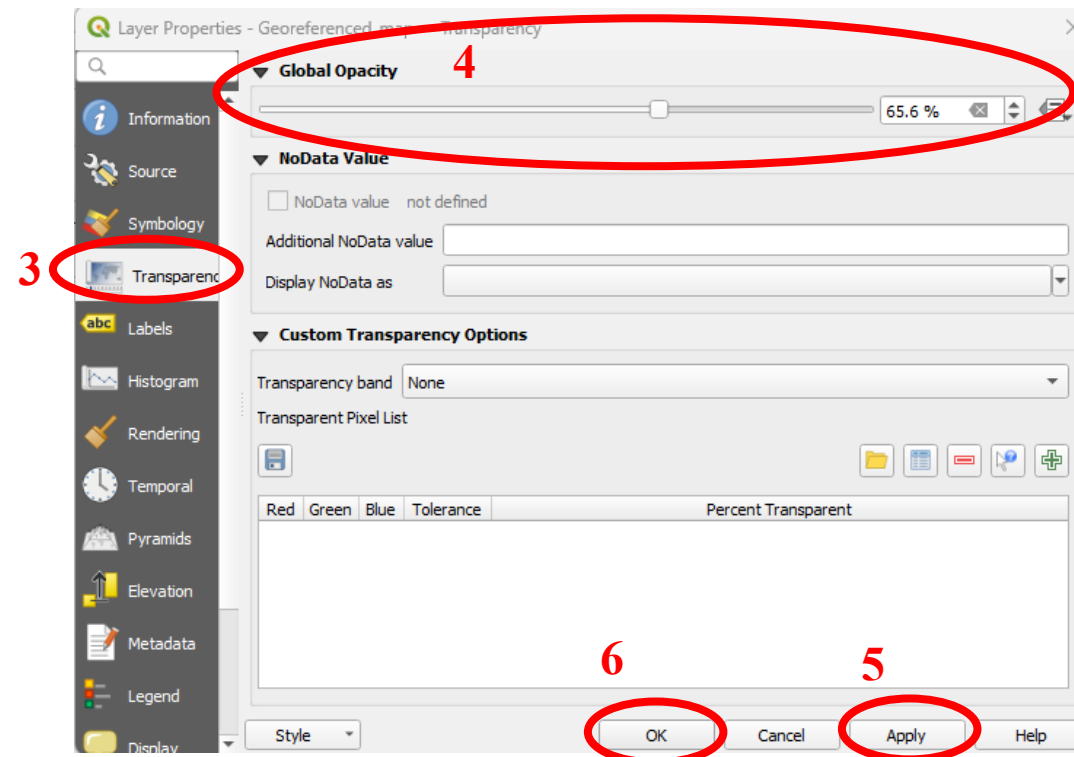
Activity #4



Change transparency

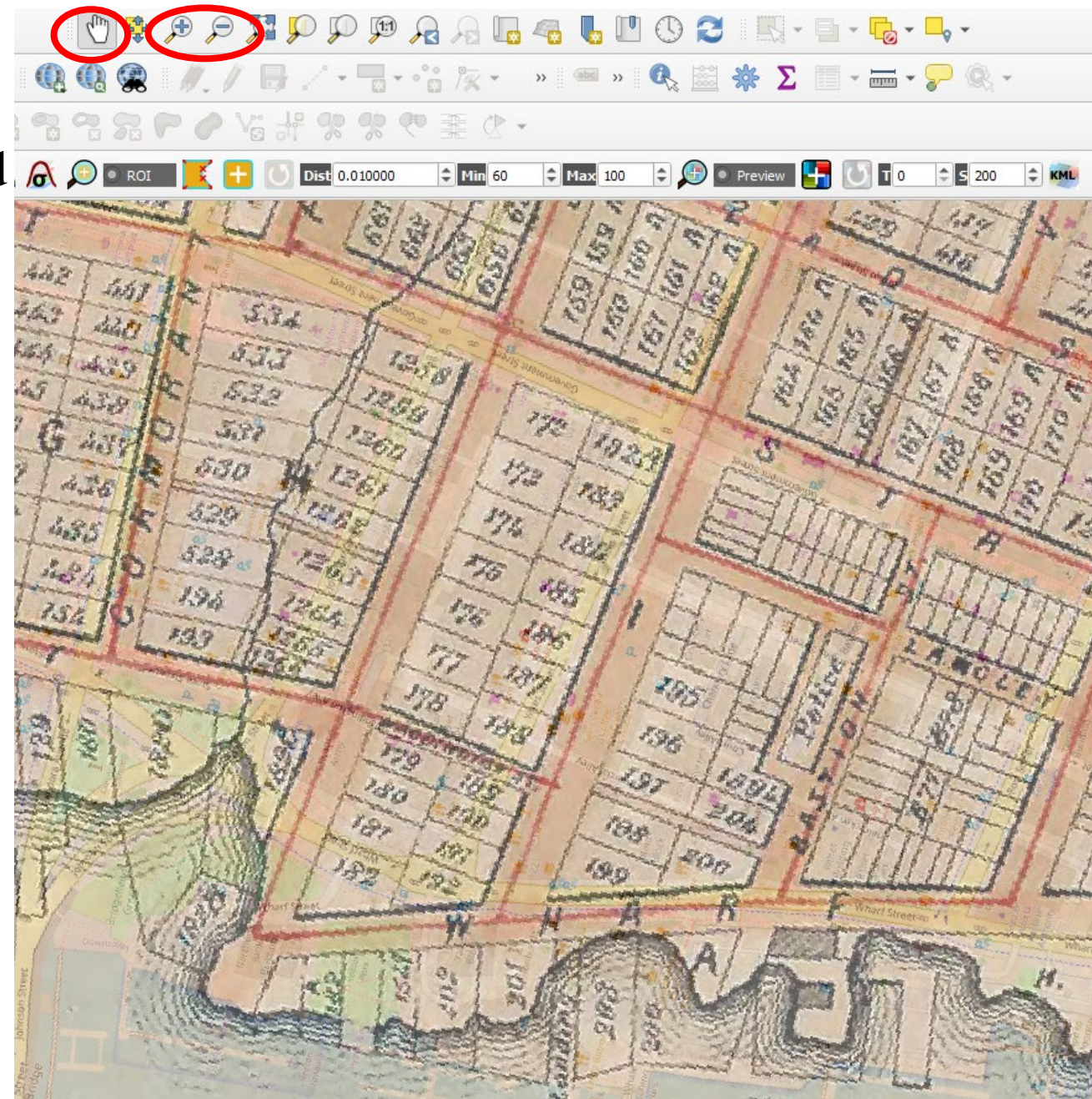


- Check accuracy of georeferencing by changing transparency of map
- Right-click **Georeferenced_map** in *Layers* and select *Properties*
- Select *Transparency*
- Use sliding scale to adjust Global Opacity and click *Apply*
- When you can see enough of the basemap underneath to check accuracy, click *OK*

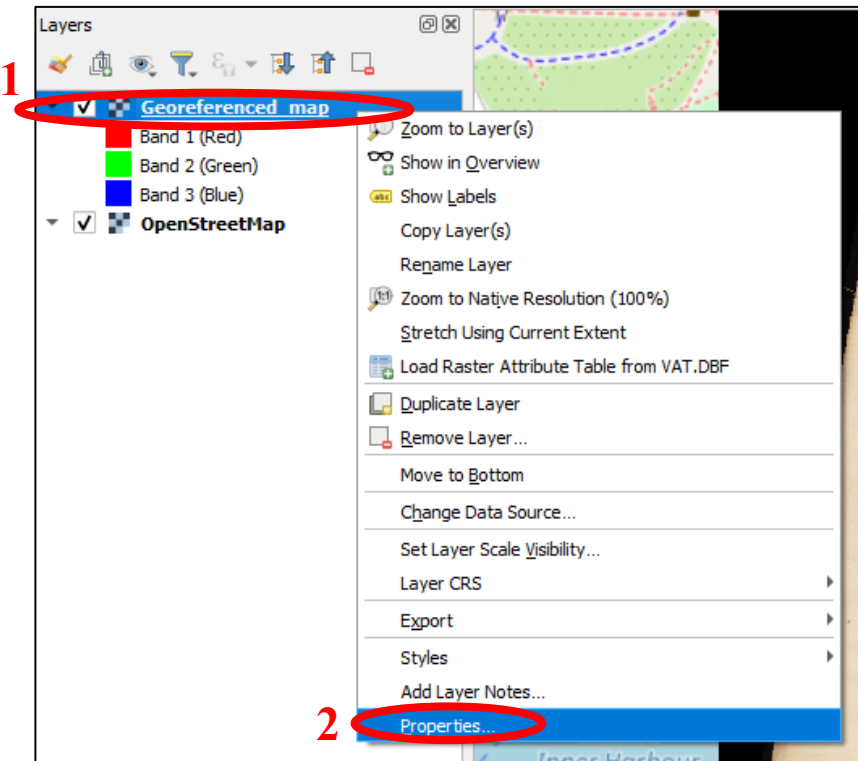


Check georeferenced accuracy

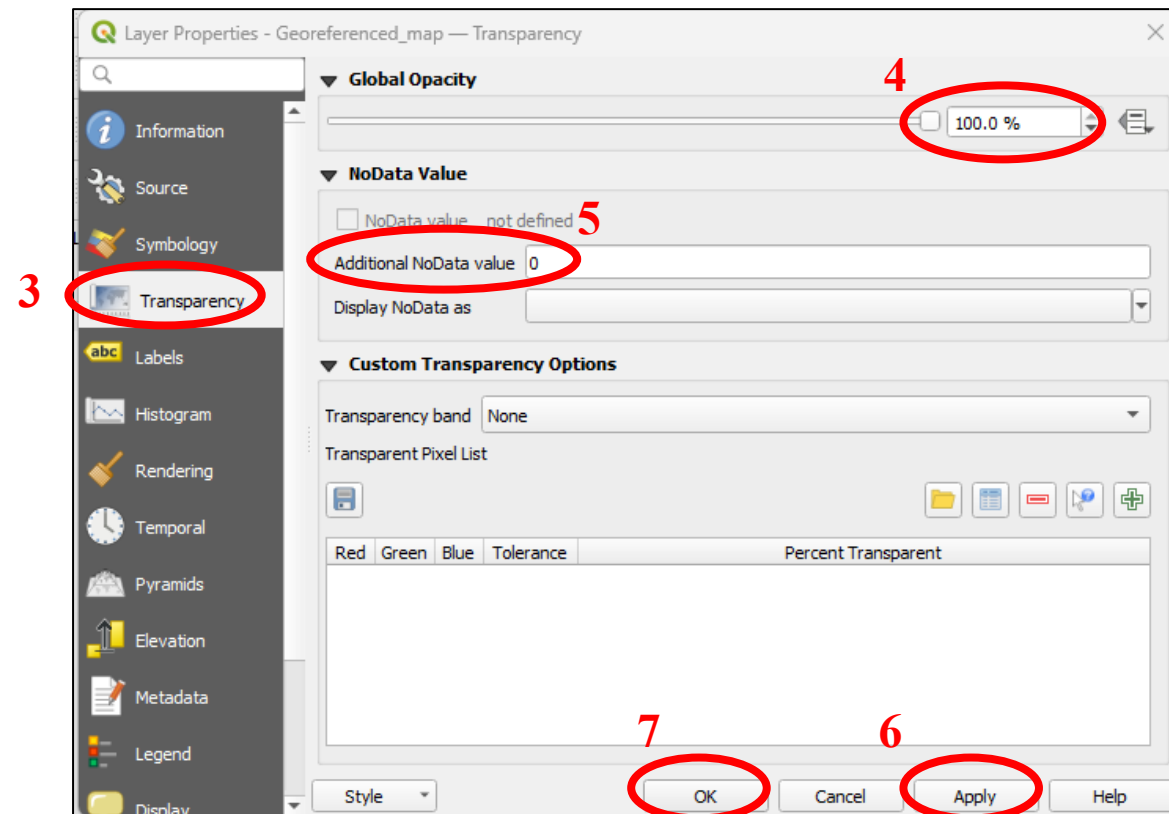
- Use Pan and Zoom to move around the Map View to see how well the map was georeferenced.
- Do the streets and other features line up?



Change transparency



- Can remove black pixels around map to it look ‘better’
- Right-click again on **Georeferenced_map** and then *Properties* and *Transparency*
- Set *Global Opacity* back to 100%
- Type “0” into box beside “Additional NoData value” to indicate that any black or white pixels should be made transparent
- *Apply* and *OK*



CHECK IN #4

Save your work!

- Checked georeferencing accuracy
- Set NoData values to transparent

If you have questions, **ask!**



Congratulations!

You can:

- Use georeferencing tool to georeference a scanned map
- Use an existing open data source as target data to georeference to



Resources going forward:



QGIS – used in workshop today:

- QGIS Tutorials & Tips: <https://www.qgistutorials.com/>
- QGIS Training Manual: https://docs.qgis.org/3.40/en/docs/training_manual/index.html
- QGIS User Guide: https://docs.qgis.org/3.40/en/docs/user_manual/index.html

Find data:

- GeoSpatial Data Guide: <http://libguides.uvic.ca/geospatialdata>

Questions or problems:

- UVic Geospatial Librarian (danielbm@uvic.ca), YCW Geospatial Intern (gabriellewade@uvic.ca), or KULA Geospatial Assistant (jeronomo.elenes@gmail.com)



UVic full semester GIS courses in the Department of Geography:

- GEOG222 – Intro to Maps and GIS
- GEOG328 – GIS Analysis

GIS Skills and Mapping Micro-certificate (36 hours)

- <https://continuingstudies.uvic.ca/science-and-the-environment/programs/gis-skills-and-mapping>